Rows → outer loop

Columns → Inner loop

Q.1. Print an nxn Star Square from user.

→
```
import java.util.Scanner;
class Demo {
        psvm() {
            Scanner sc = new Scanner(System.in);
            soPrint("Enter no. of rows: ");
            int n = sc.nextInt();
            for (int i = 1; i<=n; i++) {
                for (int j = 1; j<=n; j++) {
                    soPrint("* ");
                }
            }  sop();
        }
}
```

o/p - Enter no. of rows : 4

```
* * * * → rows
* * * *
* * * *
* * * *
↓
vertically
(column)
```

° Dry Run -

n = 4 i ≤ n

int i = 1 i ≤ 4 i++ i++

1 ≤ 4

↓

j = 1 j ≤ 4 j++

1 ≤ 4

↳ * 2

2 ≤ 4

↳ * 3

3 ≤ 4

↳ * 4

4 ≤ 4

↳ * 5

4 ≤ 5

× j loop exit

Scanned with OKEN Scanner

$2 \leq 4$

↳ $j = 1$ $\quad j \leq 4 \quad j++$

$1 \leq 4$

↳ ★ 2

$2 \leq 4$

↳ ★ 3

$3 \leq 4$

↳ ★ 4

$4 \leq 4$

↳ ★ 5

$5 \leq 4$

× (loop exit)

3

$3 \leq 4$

↳ $j = 1$ $\quad j \leq 4 \quad j++$

$1 \leq 4$

↳ ★ 2

$2 \leq 4$

↳ ★ 3

$3 \leq 4$

↳ ★ 4

$4 \leq 4$

↳ ★ 5

$5 \leq 4$

× (loop exit)

4

$4 \leq 4$

↳ $j=1$ $\quad j \leq 4 \quad j++$

$1 \leq 4$

↳ ★ 2

$2 \leq 4$

↳ ★ 3

$3 \leq 4$

↳ ★ 4

$4 \leq 4$

↳ ★ 5

$5 \leq 4$

× (loop exist)

$5 \leq 4$

Q.2. Print

```
1 1 1 1
2 2 2 2
3 3 3 3
4 4 4 4
```

→

```
import java.util.Scanner;
class Demo {
    psvm() {
        Scanner sc = new Scanner(System.in);
            sop("Enter no. of rows : ");
            int n = sc.nextInt();
            for (int i=1; i<=n; i++) {
                for (int j=1; j≤n; j++) {
                    sop(i+" ");
                }
                sop();
            }
    }
}
```

o/p- Enter no. of rows : 4

```
1 1 1 1
2 2 2 2
3 3 3 3
4 4 4 4
```

• Dry Run -

n=4

int i=1 i≤n

1≤4 i++

↳ j=1 j≤4 j++

1≤4 ↳ 1 2

2≤4 ↳ 2 3

3≤4 ↳ 3 4

4≤4 ↳ 4 5

5≤4 ×

i++

$2 \leq 4$

↳ j = 1 j ≤ 4 j++

$1 \leq 4$

↳ 1 2

$2 \leq 4$

↳ 2 3

$3 \leq 4$

↳ 3 4

$4 \leq 4$

↳ 4 5

$5 \leq 4$

X

3

$3 \leq 4$

↳ j=1 j ≤ 4 j++

$1 \leq 4$

↳ 1 2

$2 \leq 4$

↳ 2 3

$3 \leq 4$

↳ 3 4

$4 \leq 4$

↳ 4 5

$5 \leq 4$

X

4

$4 \leq 4$

↳ j=1 j ≤ 4 j++

$1 \leq 4$

↳ 1 2

$2 \leq 4$

↳ 2 3

$3 \leq 4$

↳ 3 4

$4 \leq 4$

↳ 4 5

$5 \leq 4$

X

$5 \leq 4$ X (loop exit)

5

Q.3. Print

```
1 2 3 4
1 2 3 4
1 2 3 4
1 2 3 4
```

```
import java.util.Scanner;
class Demo {
        psvm() {
            Scanner sc = new Scanner(System.in);
            sop("Enter no. of column:");
            int n = sc.nextInt();
            for (int i = 1; i<=n; i++) {
                for (int j = 1; j<=n; j++) {
                    sop(j+" ");
                }
            sop();
            }
        }
}
```

o/p - Enter no. of Column : 4

```
1 2 3 4
1 2 3 4
1 2 3 4
1 2 3 4
```

• Dry Run -

n = 4

int i = 1 i<=n

1 <= 4 i++

↳ j = 1 j<=4 j++

1 <= 4

↳ 1 2

2 <= 4

↳ 2 3

3 <= 4

↳ 3 4

4 <= 4

↳ 4 5

5 <= 4

↳ ✗

i++

$2 \leq 4$

↳ j = 1 $\quad j \leq 4 \quad j++$

$1 \leq 4$

↳ 1 $\quad 2$

$2 \leq 4$

↳ 2 $\quad 3$

$3 \leq 4$

↳ 3 $\quad 4$

$4 \leq 4$

↳ 4 $\quad 5$

$5 \leq 4$

X

3

$3 \leq 4$

↳ j = 1 $\quad j \leq 4 \quad j++$

$1 \leq 4$

↳ 1 $\quad 2$

$2 \leq 4$

↳ 2 $\quad 3$

$3 \leq 4$

↳ 3 $\quad 4$

$4 \leq 4$

↳ 4 $\quad 5$

$5 \leq 4$

X

4

$4 \leq 4$

↳ j = 1 $\quad j \leq 4 \quad j++$

$1 \leq 4$

↳ 1 $\quad 2$

$2 \leq 4$

↳ 2 $\quad 3$

$3 \leq 4$

↳ 3 $\quad 4$

$4 \leq 4$

↳ 4 $\quad 5$

$5 \leq 4$ X

$5 \leq 4$ X